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S E 417
HW3
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Question 1:

Hand in the first and last 5 test frames of this file and answer the following questions

Test Case 1 (Key = 1.1.1.1.1.1.)

help : on
version : on
order : numeric
check : yes
unique : yes
file_contents : sorted_alphabetically
duplicates : yes

Test Case 2 (Key = 1.1.1.1.1.2.)

help : on
version : on
order : numeric
check : yes
unique : yes
file_contents : sorted_alphabetically
duplicates : no

Test Case 3 (Key = 1.1.1.1.2.1.)

help : on
version : on
order : numeric
check : yes
unique : yes
file_contents : sorted_numerically
duplicates : yes

Test Case 4 (Key = 1.1.1.1.2.2.)

help : on
version : on
order : numeric

check : yes
unique : yes
file_contents : sorted_numerically
duplicates : no

Test Case 5 (Key = 1.1.1.1.1.3.1.)

help : on
version : on
order : numeric
check : yes
unique : yes
file_contents : unsorted
duplicates : yes

...

Test Case 380 (Key = 2.2.4.2.2.1.2.)

help : off
version : off
order : reverse
check : no
unique : no
file_contents : sorted_alphabetically
duplicates : no

Test Case 381 (Key = 2.2.4.2.2.2.1.)

help : off
version : off
order : reverse
check : no
unique : no
file_contents : sorted_numerically
duplicates : yes

Test Case 382 (Key = 2.2.4.2.2.2.2.)

help : off
version : off
order : reverse
check : no
unique : no

file_contents : sorted_numerically
duplicates : no

Test Case 383 (Key = 2.2.4.2.2.3.1.)

help : off
version : off
order : reverse
check : no
unique : no
file_contents : unsorted
duplicates : yes

Test Case 384 (Key = 2.2.4.2.2.3.2.)

help : off
version : off
order : reverse
check : no
unique : no
file_contents : unsorted
duplicates : no

(a) Pick one test frame and explain (on paper) what this frame means (i.e. what combinations of options you would test based on the frame – note you will want to choose one that does not have the help and/or the version turned on for this question). Use the existing input files to pick a file that would make sense to use for this test frame. Give this frame as a concrete test case and show a screen shot of the output (you will need to show the command line so we know which input you used).

Test Case 310 (Key = 2.2.1.2.2.2.2.)

help : off
version : off
order : numeric
check : no
unique : no
file_contents : sorted_numerically
duplicates : no

This test case ensures that the numeric sort works on a file that is already sorted numerically and has no duplicates. Essentially, this is a sanity check. check: no and unique: no means there is no -c or -u in the command.

```
nathan-krieger ➤ main ➤ sort -n sorted-numbers-unique.txt
3
4
10
90
101
345
455
900
```

(b) Now run the tsl on the file called “sort.refined.tsl”. How does sort.refined.tsl modify the tests cases that used help and version. Explain the differences and list how many frames are in the refined file

```
nathan-krieger ➤ main ➤ ./tsl sort.refined.tsl

98 test frames generated and written to sort.refined.tsl.tsl
```

Sort refined tsl is different because it removes the redundant tests containing help/version. Since those two are meant to be exclusive (meaning if help is on the other flags don't matter, if version is on, the other flags don't matter) there is no need to test every combination with them. The sort refined tsl generated 98 test frames.

Question 2:

(a) How many combinations (test cases) are there for each of the above files?

./cit_generate input.txt -F
9 combinations

./cit_generate cit-sort-input.txt -F
12 combinations

./cit_generate cit-sort-input-3way.txt -F
24 combinations

(b) Provide the output from the first one (cit-sort-input.txt)

```
nathan-krieger > main > cat cit_sample.out\  
>  
t   k   v   TCount  
2 4 11 45  
  
3 3 3 2  
  
0 1 2  
3 4 5  
6 7 8  
9 10  
  
9  
1 3 6 10  
0 3 7 9  
1 4 8 9  
2 4 7 10  
1 5 7 9  
2 5 6 9  
0 4 6 9  
2 3 8 9  
0 5 8 10
```

(c) (Write your own program) or manually map the test cases from cit-sort-input.txt so to use the choices from the TSL.

```
nathan-krieger > main ./cit_generate input.txt -F
first part of exp table
1 0.367879 0.135335 0.0497871 0.0183156 0.00673795 0.00247875 0.000911882 0.000335463 0.00012341 4.53999e-
05 1.67017e-05 6.14421e-06 2.26033e-06 8.31529e-07 3.05902e-07 1.12535e-07 4.13994e-08 1.523e-08 5.6028e-09
2.06115e-09 7.58256e-10 2.78947e-10 1.02619e-10 3.77513e-11 1.38879e-11 5.10909e-12 1.87953e-12 6.9144e-13
2.54367e-13 9.35762e-14 3.44248e-14 1.26642e-14 4.65889e-15 1.71391e-15 6.30512e-16 2.31952e-16 8.53305e-1
7 3.13913e-17 1.15482e-17
kcombo
8 8 8 9
Looking for cover:
t k v TCount
2 4 11 45

3 3 3 2

0 1 2
3 4 5
6 7 8
9 10

upperbound = 10009
current tabsize 4089
current tabsize 2048
current tabsize 1028
current tabsize 518
current tabsize 263
current tabsize 135
current tabsize 71
current tabsize 39
current tabsize 23
current tabsize 15
current tabsize 11
current tabsize 9
```

cit_sample.txt

There are two input files for sort:

cit-sort-input.txt

cit-sort-input-3way.txt

These models each have 5 parameters (they match the TSL but leave out the "version" parameters.

Try running these (note if you use the -F option you will overwrite the cit-sample.out file where ##### is a random number):

- > ./cit_generate cit-sort-input.txt -F
- > ./cit_generate cit-sort-input-3way.txt -F

Question 2:

(a) How many combinations (test cases) are there for each of the above files?

(c) (Write your own program) or manually map the test cases from cit-sort-input.txt so that they use the choices from the TSL.

Cit-sort-input.txt:

```
[~/git/SE417/HW3/CIT_Tool]
nathan-krieger main cat cit-sort-input.txt
2
5
4 1
2 2
3 1
2 1
```

Mapping:

Combination strength is 2

5 parameters

The first param has 4 options

The next 2 have 2 options

The 4th parameter has 2 options

The 5th parameter has 1 option

	Switch category	Numbers	Mapping
4 1	order	0 1 2 3	0 = numeric 1 = default 2 = random 3 = reverse
2 2	check	4 5	4 = yes 5 = no
2 2	unique	6 7	6 = yes 7 = no
3 1	File contents	8 9 10	8 = alpha 9 = numeric 10 = unsorted
2 1	duplicates	11 12	11 = yes 12 = no

The 12 test cases:

1 4 6 8 11

1 5 6 10 11

2 5 7 10 11

1 5 7 9 12
3 5 6 10 11
3 5 6 9 12
2 5 6 9 12
3 4 7 8 12
0 5 7 8 11
2 4 6 8 11
0 4 6 9 11
0 4 7 10 12

Mapped tests:

Line 1: 1 4 6 8 11

1: Order = default(alpha)

4: Check = yes

6: Unique = yes

8: File Contents = sorted_alphabetically

11: Duplicates = yes

Mapped Test: sort -c -u sorted.words.txt

Line 2: 1 5 6 10 11

1: Order = default(alpha)

5: Check = no

6: Unique = yes

10: File Contents = unsorted

11: Duplicates = yes

Mapped Test: sort -u random-numbers.txt

Line 3: 2 5 7 10 11

2: Order = random

5: Check = no

7: Unique = no

10: File Contents = unsorted

11: Duplicates = yes

Mapped Test: sort -R random-numbers.txt

Line 4: 1 5 7 9 12

1: Order = default(alpha)

5: Check = no

7: Unique = no

9: File Contents = sorted_numerically

12: Duplicates = no

Mapped Test: sort sorted-numbers-unique.txt

Line 5: 3 5 6 10 11
3: Order = reverse
5: Check = no
6: Unique = yes
10: File Contents = unsorted
11: Duplicates = yes
Mapped Test: sort -r -u random-numbers.txt

Line 6: 3 5 6 9 12
3: Order = reverse
5: Check = no
6: Unique = yes
9: File Contents = sorted_numerically
12: Duplicates = no
Mapped Test: sort -r -u sorted-numbers-unique.txt

Line 7: 2 5 6 9 12
2: Order = random
5: Check = no
6: Unique = yes
9: File Contents = sorted_numerically
12: Duplicates = no
Mapped Test: sort -R -u sorted-numbers-unique.txt

Line 8: 3 4 7 8 12
3: Order = reverse
4: Check = yes
7: Unique = no
8: File Contents = sorted_alphabetically
12: Duplicates = no
Mapped Test: sort -r -c sorted.words.txt

Line 9: 0 5 7 8 11
0: Order = numeric
5: Check = no
7: Unique = no
8: File Contents = sorted_alphabetically
11: Duplicates = yes
Mapped Test: sort -n sorted.words.txt

Line 10: 2 4 6 8 11
2: Order = random
4: Check = yes
6: Unique = yes

8: File Contents = sorted_alphabetically
11: Duplicates = yes
Mapped Test: sort -R -c -u sorted.words.txt

Line 11: 0 4 6 9 11
0: Order = numeric
4: Check = yes
6: Unique = yes
9: File Contents = sorted_numerically
11: Duplicates = yes
Mapped Test: sort -n -c -u sorted-numbers-dups.txt

Line 12: 0 4 7 10 12
0: Order = numeric
4: Check = yes
7: Unique = no
10: File Contents = unsorted
12: Duplicates = no
Mapped Test: sort -n -c random-numbers.unique.txt

Question 3:

```
nathan-krieger > main > ./tsl firefox.tsl  
1792 test frames generated and written to firefox.tsl.tsl
```

(a) How many test frames do you have in this file?

1792

(b) Copy the contents of the firefox.tsl to the document. Copy the first and last 5 tests (10 in total) of the firefox.tsl.tsl file to this document

```

1  #tsl for sort
2  Parameters:
3      search_engine:
4          Google.
5          Amazon.
6          Bing.
7          DuckDuckGo.
8          eBay.
9          Perplexity.
10         Wikipedia.
11     tab_opening_links_in_tabs:
12         on.
13         off.
14     tab_opening_switch_to_link:
15         on.
16         off.
17     tab_opening_links_next_to_active_tab:
18         on.
19         off.
20     tab_interaction_ctrl_tab_recently_used:
21         on.
22         off.
23     tab_interaction_image_preview:
24         on.
25         off.
26     browser_layout:
27         horizontal_tabs.
28         vertical_tabs.
29     firefox_suggest_history:
30         on.
31         off.
32     firefox_suggest_bookmarks:
33         on.
34         off.
35

```

Test Case 1 (Key = 1.1.1.1.1.1.1.1.)

search_engine : Google
tab_opening_links_in_tabs : on
tab_opening_switch_to_link : on
tab_opening_links_next_to_active_tab : on
tab_interaction_ctrl_tab_recently_used : on
tab_interaction_image_preview : on
browser_layout : horizontal_tabs

firefox_suggest_history : on
firefox_suggest_bookmarks : on

Test Case 2 (Key = 1.1.1.1.1.1.1.2.)

search_engine : Google
tab_opening_links_in_tabs : on
tab_opening_switch_to_link : on
tab_opening_links_next_to_active_tab : on
tab_interaction_ctrl_tab_recently_used : on
tab_interaction_image_preview : on
browser_layout : horizontal_tabs
firefox_suggest_history : on
firefox_suggest_bookmarks : off

Test Case 3 (Key = 1.1.1.1.1.1.2.1.)

search_engine : Google
tab_opening_links_in_tabs : on
tab_opening_switch_to_link : on
tab_opening_links_next_to_active_tab : on
tab_interaction_ctrl_tab_recently_used : on
tab_interaction_image_preview : on
browser_layout : horizontal_tabs
firefox_suggest_history : off
firefox_suggest_bookmarks : on

Test Case 4 (Key = 1.1.1.1.1.1.2.2.)

search_engine : Google
tab_opening_links_in_tabs : on
tab_opening_switch_to_link : on
tab_opening_links_next_to_active_tab : on
tab_interaction_ctrl_tab_recently_used : on
tab_interaction_image_preview : on
browser_layout : horizontal_tabs
firefox_suggest_history : off
firefox_suggest_bookmarks : off

Test Case 5 (Key = 1.1.1.1.1.2.1.1.)

search_engine : Google
tab_opening_links_in_tabs : on
tab_opening_switch_to_link : on
tab_opening_links_next_to_active_tab : on
tab_interaction_ctrl_tab_recently_used : on

tab_interaction_image_preview : on
browser_layout : vertical_tabs
firefox_suggest_history : on
firefox_suggest_bookmarks : on

Test Case 1788 (Key = 7.2.2.2.2.1.2.2.)

search_engine : Wikipedia
tab_opening_links_in_tabs : off
tab_opening_switch_to_link : off
tab_opening_links_next_to_active_tab : off
tab_interaction_ctrl_tab_recently_used : off
tab_interaction_image_preview : off
browser_layout : horizontal_tabs
firefox_suggest_history : off
firefox_suggest_bookmarks : off

Test Case 1789 (Key = 7.2.2.2.2.2.1.1.)

search_engine : Wikipedia
tab_opening_links_in_tabs : off
tab_opening_switch_to_link : off
tab_opening_links_next_to_active_tab : off
tab_interaction_ctrl_tab_recently_used : off
tab_interaction_image_preview : off
browser_layout : vertical_tabs
firefox_suggest_history : on
firefox_suggest_bookmarks : on

Test Case 1790 (Key = 7.2.2.2.2.2.1.2.)

search_engine : Wikipedia
tab_opening_links_in_tabs : off
tab_opening_switch_to_link : off
tab_opening_links_next_to_active_tab : off
tab_interaction_ctrl_tab_recently_used : off
tab_interaction_image_preview : off
browser_layout : vertical_tabs
firefox_suggest_history : on
firefox_suggest_bookmarks : off

Test Case 1791 (Key = 7.2.2.2.2.2.2.1.)

search_engine : Wikipedia
tab_opening_links_in_tabs : off

tab_opening_switch_to_link : off
tab_opening_links_next_to_active_tab : off
tab_interaction_ctrl_tab_recently_used : off
tab_interaction_image_preview : off
browser_layout : vertical_tabs
firefox_suggest_history : off
firefox_suggest_bookmarks : on

Test Case 1792 (Key = 7.2.2.2.2.2.2.2.)

search_engine : Wikipedia
tab_opening_links_in_tabs : off
tab_opening_switch_to_link : off
tab_opening_links_next_to_active_tab : off
tab_interaction_ctrl_tab_recently_used : off
tab_interaction_image_preview : off
browser_layout : vertical_tabs
firefox_suggest_history : off
firefox_suggest_bookmarks : off

Question 4:

(a) Hand in a screen shot of both of these files (the input file and the 2-way sample).

```
nathan-krieger ➤ main ➤ cat firefox.input.txt
2
9
7 1
2 8%
```

```

nathan-krieger main cat cit_sample.out
t k v TCount
2 9 23 224

7 2 2 2 2 2 2 2

0 1 2 3 4 5 6
7 8
9 10
11 12
13 14
15 16
17 18
19 20
21 22

14
6 8 9 12 13 16 18 19 21
0 8 10 12 14 16 18 20 21
4 7 9 11 14 15 18 20 22
2 7 10 12 14 16 17 20 22
4 8 10 12 13 16 17 19 21
5 8 10 12 13 15 17 20 21
3 8 10 11 14 16 17 20 22
1 8 10 12 14 15 18 19 21
6 7 10 11 14 15 17 20 22
1 7 9 11 13 16 17 20 22
2 8 9 11 13 15 18 19 21
3 7 9 12 13 15 18 19 21
5 7 9 11 14 16 18 19 22
0 7 9 11 13 15 17 19 22

```

(b) Map the 2-way sample file to the configuration names and hand in the mapped file. This should match the file handed in in part a.

Integers	Parameter	Mapping
0 - 6	Search Engine	0=Google, 1=Amazon, 2=Bing, 3=DuckDuckGo, 4=eBay, 5=Perplexity, 6=Wikipedia
7, 8	Tab: Opening Links	7=on, 8=off
9, 10	Tab: Switch to Link	9=on, 10=off
11, 12	Tab: Next to Active	11=on, 12=off
13, 14	Tab: Ctrl+Tab	13=on, 14=off
15, 16	Tab: Image Preview	15=on, 16=off
17, 18	Browser Layout	17=Horizontal, 18=Vertical
19, 20	Suggest: History	19=on, 20=off
21, 22	Suggest: Bookmarks	21=on, 22=off

1. 6 8 9 12 13 16 18 19 21 Wikipedia, Tab-Links-Off, Switch-Link-On, Next-Active-Off, Ctrl-Tab-On, Preview-Off, Vertical-Layout, History-On, Bookmarks-On

2. 0 8 10 12 14 16 18 20 21 Google, Tab-Links-Off, Switch-Link-Off, Next-Active-Off, Ctrl-Tab-Off, Preview-Off, Vertical-Layout, History-Off, Bookmarks-On

3. 4 7 9 11 14 15 18 20 22 eBay, Tab-Links-On, Switch-Link-On, Next-Active-On, Ctrl-Tab-Off, Preview-On, Vertical-Layout, History-Off, Bookmarks-Off

4. 2 7 10 12 14 16 17 20 22 Bing, Tab-Links-On, Switch-Link-Off, Next-Active-Off, Ctrl-Tab-Off, Preview-Off, Horizontal-Layout, History-Off, Bookmarks-Off

5. 4 8 10 12 13 16 17 19 21 eBay, Tab-Links-Off, Switch-Link-Off, Next-Active-Off, Ctrl-Tab-On, Preview-Off, Horizontal-Layout, History-On, Bookmarks-On

6. 5 8 10 12 13 15 17 20 21 Perplexity, Tab-Links-Off, Switch-Link-Off, Next-Active-Off, Ctrl-Tab-On, Preview-On, Horizontal-Layout, History-Off, Bookmarks-On

7. 3 8 10 11 14 16 17 20 22 DuckDuckGo, Tab-Links-Off, Switch-Link-Off, Next-Active-On, Ctrl-Tab-Off, Preview-Off, Horizontal-Layout, History-Off, Bookmarks-Off

8. 1 8 10 12 14 15 18 19 21 Amazon, Tab-Links-Off, Switch-Link-Off, Next-Active-Off, Ctrl-Tab-Off, Preview-On, Vertical-Layout, History-On, Bookmarks-On

9. 6 7 10 11 14 15 17 20 22 Wikipedia, Tab-Links-On, Switch-Link-Off, Next-Active-On, Ctrl-Tab-Off, Preview-On, Horizontal-Layout, History-Off, Bookmarks-Off

10. 1 7 9 11 13 16 17 20 22 Amazon, Tab-Links-On, Switch-Link-On, Next-Active-On, Ctrl-Tab-On, Preview-Off, Horizontal-Layout, History-Off, Bookmarks-Off

11. 2 8 9 11 13 15 18 19 21 Bing, Tab-Links-Off, Switch-Link-On, Next-Active-On, Ctrl-Tab-On, Preview-On, Vertical-Layout, History-On, Bookmarks-On

12. 3 7 9 12 13 15 18 19 21 DuckDuckGo, Tab-Links-On, Switch-Link-On, Next-Active-Off, Ctrl-Tab-On, Preview-On, Vertical-Layout, History-On, Bookmarks-On

13. 5 7 9 11 14 16 18 19 22 Perplexity, Tab-Links-On, Switch-Link-On, Next-Active-On, Ctrl-Tab-Off, Preview-Off, Vertical-Layout, History-On, Bookmarks-Off

14. 0 7 9 11 13 15 17 19 22 Google, Tab-Links-On, Switch-Link-On, Next-Active-On, Ctrl-Tab-On, Preview-On, Horizontal-Layout, History-On, Bookmarks-Off

(c) Assume you have a base choice using the default values above. You can assume that Bing is the default browser.

Create a sample for Base Choice Criteria (this has to be done by hand)

Hand in the Base Choice criteria

Base choice:

Search engine: Bing

All other settings: on

Test #	Search Engine	Tab : Links	Tab: Switch	Tab : Next	Tab: Ctrl+Tab	Tab: Preview	Layout	Sug: History	Sug: Bookmarks
1	Bing	on	off	off	off	on	Horiz.	on	on
2	Google	on	off	off	off	on	Horiz.	on	on
3	Amazon	on	off	off	off	on	Horiz.	on	on
4	Duck Duck Go	on	off	off	off	on	Horiz.	on	on
5	eBay	on	off	off	off	on	Horiz.	on	on
6	Perplexity	on	off	off	off	on	Horiz.	on	on
7	Wikipedia	on	off	off	off	on	Horiz.	on	on

8	Bing	off	off	off	off	on	Horiz.	on	on
9	Bing	on	on	off	off	on	Horiz.	on	on
10	Bing	on	off	on	off	on	Horiz.	on	on
11	Bing	on	off	off	on	on	Horiz.	on	on
12	Bing	on	off	off	off	off	Horiz.	on	on
13	Bing	on	off	off	off	on	Vert.	on	on
14	Bing	on	off	off	off	on	Horiz.	off	on
15	Bing	on	off	off	off	on	Horiz.	on	off